

Sensor Technologies for Wastewater Collection Networks: Characteristics, Applications, and Limitations

Abstract

Wastewater collection networks can be monitored for water level, flow, water quality, gas conditions, and pipe condition. Each purpose requires a different sensor or inspection method. This review compares the main technologies used in sewer networks, with emphasis on their practical use, strengths, and limitations. Non-contact radar and ultrasonic sensors are widely applicable to level monitoring, although their performance depends on installation and environmental conditions. Flow estimates require both hydraulic context and suitable calibration. Water-quality probes, UV/Vis instruments, and fluorescence sensors can identify changes in wastewater composition, but fouling and matrix effects remain important. Hydrogen sulfide monitoring also requires a clear distinction between gas-phase and liquid-phase measurements. The review shows that sensor selection should begin with the operational question and the conditions at the intended site.

Keywords: wastewater collection; sewer monitoring; level sensing; flow sensing; water quality; optical spectroscopy; hydrogen sulfide; operational monitoring.

1 Introduction

Sewer monitoring is used to detect surcharge and overflow risk, estimate hydraulic loading, identify unusual discharges, manage odour and corrosion, and assess pipe condition. These tasks do not produce the same type of information. A level sensor records hydraulic behaviour, a water-quality probe responds to changes in wastewater composition, and an inspection system examines the physical condition of the pipe. Telemetry then carries the measurements from the sewer to an operator or data platform.

The useful characteristics of a sensor depend on its location and purpose. Humidity, condensation, foam, solids, corrosion, limited access, radio coverage, and power availability can all influence performance. The following sections group sensor technologies by the practical question they answer and compare their applications and limitations.

2 Monitoring Questions and Sensor Categories

The literature is clearer when grouped by the question being answered rather than by a single list of hardware names.

Table 1: Question-led classification of sewer monitoring technologies.

Question	Examples	Information obtained	Main limitation	Sources (Ref., pp.)
How full is the sewer?	Radar, ultrasonic, short-range LiDAR/ToF, float or tilt switches	Continuous level trends or discrete surcharge events	Humidity, condensation, foam, obstructions, fouling, or loss of trend information	[6, pp. 3–4, 28–29]; [5, pp. 1–3, 6–9]; [21, pp. 7, 10]; [20, pp. 3–1–3–4]
How much is flowing?	Contactless ultrasonic flow, level plus geometry, pump drawdown or hydraulic models	Flow or velocity estimates and hydraulic loading	Installation geometry, flow direction, backwater, and calibration assumptions	[1, pp. 1–4, 11–18]; [6, pp. 13, 23–24]
What is in the wastewater?	pH, conductivity, temperature, turbidity, UV/Vis, fluorescence, H ₂ S	Chemical changes, illicit-discharge indicators, organic-load proxies, odour or corrosion indicators	Fouling, matrix effects, local calibration, and phase-specific measurement	[16, pp. 2–4, 11–12]; [9, pp. 1, 3–4, 8–9]; [8, pp. 1–2, 10–11]; [3, pp. 1, 5–6, 12]; [12, pp. 1–5, 10–15]; [4, pp. 13–16]
Is the asset blocked or damaged?	Acoustic screening, CCTV, computer vision, fibre-optic or other condition methods	Structural or blockage information	Usually an inspection or diagnostic workflow rather than a continuous level time series	[2, pp. 1–5]; [22, pp. 1–8]; [15, pp. 1–4, 9–12]; [11, pp. 1–6]; [14, pp. 1–6]
How does the reading get back?	LoRaWAN, cellular IoT, gateways, data platforms, soft sensors	Remote transmission and estimation of unmeasured states	Coverage, power, data quality, model dependence, and site-specific validation	[6, pp. 15–19]; [17, pp. 1–6, 12–15]; [10, pp. 1–7, 13–18]

Telemetry and soft sensors are enabling layers rather than physical hydraulic or water-quality sensors. Acoustic and CCTV methods are also distinct from fixed probes because they examine asset condition instead of producing a continuous level or chemical measurement.

3 Level and Depth Sensing

Continuous level data provide a time series of depth from which a utility may examine normal patterns, surcharge, and rate of change. A binary float or tilt switch provides an event threshold but cannot produce the same trend information. The choice therefore depends on whether the monitoring purpose is broad event notification, hydraulic diagnosis, or both.

3.1 Non-contact Radar

Radar level sensors use an electromagnetic signal reflected from the wastewater surface to measure the distance between the sensor and the water. Because the sensor is mounted above the flow, it avoids direct contact with wastewater. Its performance still depends on the selected instrument, mounting position, beam geometry, and site conditions.

Drenoyanis et al. [6] used non-contact radar for sewer level measurement and pumping-station drawdown tests. The study describes a portable drawdown device and a LoRaWAN-enabled prototype installed in the Shoalhaven Water network. A separate radio survey recorded communication distances of up to 20 km in an urban area and up to 35 km under line-of-sight conditions. These distances describe the LoRaWAN link, not the radar measurement range.

Radar was less affected by humidity and mist than ultrasonic sensing in the comparison reported by Drenoyanis et al. Steam, condensation, foam, and changes in air temperature were identified as potential sources of unstable ultrasonic readings. Radar nevertheless requires careful installation because an unsuitable mounting arrangement can produce false detections. The study also identifies its higher cost as a disadvantage [6].

Drenoyanis et al. [6, pp. 28–29] describe an integrated 80 GHz short-range radar network with approximately five devices around each pumping station as a future development. Instrument frequency, accuracy, service life, and power requirements therefore need to be considered for the particular radar unit selected.

3.2 Ultrasonic, LiDAR, and Time-of-Flight Sensors

An ultrasonic level sensor estimates distance from the travel time of an acoustic pulse reflected by the wastewater surface. It is a common non-contact option and can provide a continuous level record at relatively low power.

Steam, mist, condensation, foam, and temperature-dependent changes in sound speed can cause unstable readings [6, pp. 3–4]. Installation must also provide a clear acoustic path between the sensor and the wastewater surface.

Do et al. [5] used continuous ultrasonic level data to detect developing sewer chokes in Stonyfell, South Australia. Their case study examined 62 datasets collected over one year and showed how changes in the normal level pattern can provide an earlier indication of a blockage.

Utepov et al. [21] compared Sharp infrared, TF-Luna LiDAR, TOF400 laser, and JSN-SR04T ultrasonic sensors in a prototype designed to imitate a sewer well. In that arrangement, the JSN-SR04T measured to approximately ± 5 cm at distances up to 2 m, while the TF-Luna result was within approximately ± 8 cm at distances up to 5 m. The reported material cost of the complete prototype was about USD 170.07, excluding labour. The tests were conducted in an imitative setup, and the authors identified trials in different sewer systems as future work.

LiDAR and other optical time-of-flight sensors offer compact distance measurement and may suit locations with a clear optical path. Their exposed optical surfaces can be affected by condensation and deposits. Field use therefore depends on protection of the sensing window, suitable mounting geometry, and testing under the conditions found in the target sewer.

3.3 Binary Float and Tilt Sentinels

Float and tilt devices answer a simpler question: has wastewater crossed a set threshold? Their switch output is useful for surcharge or overflow alerts when a continuous level record is unnecessary. A single switch cannot show the rate of rise or describe changes between its trigger and reset levels. US EPA guidance describes mechanical floats as depth sensors and notes that grease and solids can obstruct their movement in sewer applications [20]. Battery life, maintenance intervals, and cost depend on the selected switch, logger, and communication system.

4 Flow and Velocity Measurement

For a partially full sewer, discharge depends on both the wetted cross-sectional area and the mean water velocity. A level sensor alone does not automatically provide a reliable flow measurement when the flow is non-uniform or the geometry and boundary conditions are uncertain.

4.1 Contactless Ultrasonic Flow

Alshami et al. [1] studied twelve sewer locations in Hong Kong using contactless ultrasonic flow and level sensors. The level–flow correlation coefficient was 0.36, and the average relative flow-rate error was 72.14% when the sensor results were compared with Manning-equation estimates. The result reflects the instruments, sites, and comparison method used in that study.

Possible causes of error included small pipe diameters, installation angle, inconsistent flow direction, water level, and the presence of multiple inlets or outlets. In practice, contactless flow sensors need site-specific installation and calibration, making flow measurement more demanding than level measurement alone.

4.2 Level-derived Flow and Manning Models

Manning-type level-to-flow estimates can be useful when their assumptions match the target sewer reach. In a blocked or backwatered reach, the water level may rise without a proportional increase in velocity or discharge. Reliable conversion therefore depends on hydraulic context, pipe geometry, calibration, and comparison with observed flow where possible. The 72.14% result reported by Alshami et al. illustrates this problem for one network and instrument configuration.

Drenoyanis et al. [6] used radar level measurements during pumping-station drawdown to estimate pump flow indirectly. This application differs from measuring flow in a partially full gravity sewer because it relies on the changing water volume within a defined wet well.

5 Water-Quality and Chemical Monitoring

Hydraulic sensors show how full the sewer is and how its level or flow changes. Water-quality sensors provide different information: they can indicate an unusual discharge, dilution, changing organic load, or conditions associated with odour and corrosion. These sensors are most useful at locations where a chemical change has a clear operational meaning, such as an industrial inflow, a pressure-sewer discharge, or an upstream and downstream comparison point.

5.1 Physicochemical Event Detection

Shi et al. [16] deployed 22 low-cost units measuring level, temperature, and electrical conductivity across an urban Melbourne catchment for more than one year. Changes in these measurements were used to identify dry-weather illicit discharges and short-duration events. Electrical conductivity and temperature are useful event indicators, although the cause of a change still has to be interpreted from its timing, location, and surrounding conditions.

Li et al. [9] evaluated a sewage-quality monitoring system measuring temperature, conductivity, turbidity, and pH. The design included wall-jet cleaning and self-diagnostic functions. It was evaluated within a three-year field programme, and one trial site operated continuously in raw sewage for up to 24 months without maintenance or failure. This performance was achieved by the particular system and cleaning arrangement used in the study.

5.2 UV/Vis and Fluorescence

Lepot et al. [8] reviewed UV/Vis methods for estimating total suspended solids and total or dissolved chemical oxygen demand. These instruments provide rapid, reagent-free measurements, but the relationship between the optical signal and the target parameter changes with the wastewater matrix. Local calibration is therefore important when an instrument is moved to a new site or used with a different wastewater composition.

Carstea et al. [3] found that fluorescence features such as tryptophan-like Peak T and humic-like Peak C can correlate with biochemical oxygen demand, chemical oxygen demand, or total organic carbon. The strength of these relationships varies with the sample matrix and measurement method. Overlapping fluorophores, inner-filter effects, turbidity, and fouling can also alter the signal. Fluorescence is therefore most useful as a rapid calibrated proxy rather than a direct measurement of these laboratory parameters.

Optical instruments can provide frequent measurements without chemical reagents, which makes them useful for detecting short events between manual samples. Their main practical challenges are fouling, cleaning, calibration drift, and sensitivity to the local wastewater matrix. Cleaning may be provided by a wiper, flushing system, compressed air, or scheduled manual maintenance, depending on the instrument and installation.

Nutrient sensors include ion-selective and other electrochemical methods for ammonium, nitrate, and nitrite. Selectivity, interfering ions, temperature, drift, and membrane condition affect their use in complex water matrices [19]. Dissolved oxygen can be measured with amperometric or optical instruments; the two principles differ in response time, stability, flow dependence, and maintenance needs [7]. Microbial fuel-cell biosensors have also been investigated for continuous biochemical oxygen demand and toxicity monitoring, although biofilm behaviour and operating conditions influence their response [18].

6 Hydrogen Sulfide and Corrosion Monitoring

Hydrogen sulfide is relevant to sewer odour, worker safety, and biogenic corrosion. Gas-phase sensors measure concentrations in the sewer headspace, while liquid-phase sensors measure dissolved sulfide in wastewater. The two measurements have different purposes, instruments, calibration needs, and suitable installation points.

6.1 Gas-Phase Sensors

Pacheco Fernández et al. [12] compared three commercial gas sensors at two sewer sites. OdaLog and MyData devices had stated ranges of 0–200 ppm, while the SulfiLogger range was 0–1000 ppm. The reported battery specifications were approximately eight months, three months at the stated transmission configuration, and two years for the respective devices. Measurement range, logging interval, and telemetry configuration therefore need to be considered together when selecting a gas sensor.

Agreement between the gas sensors was better at low concentrations, while deviations increased as the concentration rose. High humidity and device-management requirements were also important practical considerations. A suitable gas sensor therefore needs an adequate measurement range, resistance to sewer humidity, accessible calibration, and a logging interval that captures short concentration peaks.

6.2 Liquid-Phase Sensors

Despot et al. [4] studied online liquid-phase H₂S sensors, including UV/Vis and electrochemical instruments. Local calibration, interference from solids, and regular inspection affected their operation. Useful monitoring points included the end of a pressure sewer and locations downstream of chemical dosing, where dissolved-sulfide measurements can help assess septicity and treatment performance.

Pandey et al. [13] reviewed sensor-based approaches to H₂S monitoring and their role in understanding odour and corrosion. Long-term interpretation requires attention to sensor drift, the position of the sensor,

ventilation, wastewater conditions, and the relationship between short concentration peaks and cumulative corrosion exposure.

7 Pipe Condition and Inspection Methods

Acoustic screening, CCTV, computer vision, and fibre-optic methods examine structural defects, obstructions, and changes in pipe condition [2, 11, 14, 15, 22]. Acoustic methods can screen long pipe sections and help prioritise further inspection. CCTV provides direct visual evidence but requires access, suitable flow conditions, and skilled interpretation. Computer vision can assist with consistent defect classification, although its performance depends on image quality and the data used to train the model. These methods complement continuous hydraulic monitoring rather than replacing it.

8 Fouling, Maintenance, Telemetry, and Power

Fouling, solids, condensation, calibration, and safe access influence the reliability and cost of sewer monitoring. Li et al. [9] demonstrated wall-jet cleaning for a multi-parameter water-quality system, while Despot et al. [4] found that liquid-phase H_2S sensors needed regular inspection and maintenance. Maintenance frequency depends on the sensor surface, wastewater characteristics, cleaning system, and ease of access.

Non-contact measurement avoids direct immersion and usually reduces fouling, but exposed surfaces and mounting hardware may still need inspection. Radar is sensitive to mounting and beam geometry, while ultrasonic and optical sensors can be affected by condensation or deposits. Contacting and submerged devices face greater exposure to solids, grease, and biofilm.

Drenoyanis et al. [6] provide an Australian LoRaWAN case study and regional signal survey. The recorded communication distances applied to the antenna arrangements and terrain used in that survey, so radio performance still needs to be tested at the intended monitoring sites. Sikorski et al. [17] show that sensor coverage, battery capacity, sampling frequency, and cost are linked decisions. More frequent measurement and transmission improve temporal detail but increase energy use and data volume.

Operational planning should include installation access, cleaning, calibration, communications testing, battery replacement, and data-quality checks. These requirements can outweigh the purchase price when a sensor is difficult to reach or maintain.

9 Comparison of Sensor Characteristics

Table 2: Characteristics and applications of sewer monitoring technologies.

Technology	Main use	Useful characteristics	Limitations	Sources (Ref., pp.)
Radar level	Continuous level and pump draw-down	Non-contact; relatively resistant to humidity and mist	Careful mounting and clear beam path; higher cost	[6, pp. 1, 3–4, 28–29]
Ultrasonic level	Continuous level and early surcharge detection	Non-contact; low-power options are available	Condensation, foam, temperature, and mounting conditions	[5, pp. 1–3, 6–9]; [21, pp. 7, 10]; [6, pp. 3–4]
LiDAR/ToF	Short-range distance measurement	Compact and potentially low power	Optical fouling, condensation, and limited sewer field experience	[21, pp. 7, 10, 15]
Float/tilt switch	Threshold alarm	Simple event notification	No continuous trend; movement can be obstructed by grease or solids	[20, pp. 3-1-3-4]; [17, pp. 1–6]
Contactless ultrasonic flow	Flow and level estimation	Avoids submerged installation	Sensitive to geometry, flow direction, water level, and calibration	[1, pp. 1–4, 11–18]
Physicochemical probes	Conductivity, temperature, pH, and turbidity	Detects changes in wastewater composition	Fouling, calibration drift, and interpretation of events	[16, pp. 2–4, 11–12]; [9, pp. 1, 3–4, 8–9]
UV/Vis and fluorescence	TSS and organic-load proxies	Rapid and reagent-free	Local calibration, matrix effects, and optical fouling	[8, pp. 1–2, 10–11]; [3, pp. 1, 5–6, 12]
H ₂ S sensors	Headspace gas or dissolved sulfide	Supports odour, corrosion, and dosing control	Phase-specific calibration, humidity, solids, and maintenance	[12, pp. 1–5, 10–15]; [4, pp. 13–16]; [13, pp. 1–4]
Telemetry and soft sensors	Remote data transfer and estimated states	Supports distributed monitoring and automated alerts	Coverage, power use, data quality, and model transfer	[6, pp. 15–19]; [17, pp. 1–6, 12–15]; [10, pp. 1–7, 13–18]

10 Matching Sensors to Monitoring Tasks

No single sensor describes every aspect of a wastewater network. A float or tilt switch suits a simple high-level alarm, while radar or ultrasonic sensing is more useful when the timing, rate, and duration of surcharge are important. Flow sensors are needed where hydraulic loading must be quantified. Water-quality and H₂S instruments are better suited to selected locations where a change can be linked to a discharge source, corrosion risk, or treatment action. Acoustic screening and CCTV remain appropriate for structural defects and blockages. In each case, the value of the measurement has to be considered alongside access, power, telemetry, fouling, calibration, and maintenance.

11 Conclusion

Sewer monitoring is best approached through the operational question that needs to be answered. Radar, ultrasonic, LiDAR, and float devices provide different levels of information about surcharge and water level. Radar is attractive in humid environments, while ultrasonic and optical sensors may offer lower-cost alternatives where condensation, foam, and fouling can be controlled. Flow measurement is more demanding because pipe geometry, flow direction, backwater, and calibration all influence the result.

Water-quality and optical sensors add information that hydraulic measurements cannot provide, particularly for unusual discharges, organic-load changes, odour, and corrosion. Their usefulness depends on cleaning, calibration, and the ability to interpret the signal at a particular site. Gas-phase and liquid-phase H₂S monitoring should also be selected separately because they measure different parts of the sewer environment. Overall, the most useful sensor is the one that provides enough information for a defined operational response and can be maintained reliably at the intended location.

References

- [1] Ahmad Alshami, Mohamed Elsayed, Ehab Ali, Abdelrahman E. E. Eltoukhy, and Tarek Zayed. Monitoring blockage and overflow events in small-sized sewer network using contactless flow sensors in Hong Kong: Problems, causes, and proposed solution. *IEEE Access*, 11:87131–87149, 2023. [\[Direct PDF\]](#) · doi:10.1109/ACCESS.2023.3305275.
- [2] M. T. Bin Ali, Kirill V. Horoshenkov, and Simon J. Tait. Rapid detection of sewer defects and blockages using acoustic-based instrumentation. *Water Science and Technology*, 64(8):1700–1707, 2011. [\[Direct PDF\]](#) · doi:10.2166/wst.2011.183.
- [3] E. M. Carstea, John Bridgeman, Andy Baker, and Darren M. Reynolds. Fluorescence spectroscopy for wastewater monitoring: A review. *Water Research*, 95:205–219, 2016. [\[Direct PDF\]](#) · doi:10.1016/j.watres.2016.03.021.
- [4] Danijel Despot, Mario Pacheco Fern'andez, and Matthias Barjenbruch. Comparison of online sensors for liquid phase hydrogen sulphide monitoring in sewer systems. *Water*, 13(13):1876, 2021. [\[Direct PDF\]](#) · doi:10.3390/w13131876.
- [5] Nhu Cuong Do, Luke Dix, Martin Francis Lambert, and Mark Leslie Stephens. Proactive detection of wastewater overflows for smart sanitary sewer systems: Case study in South Australia. *Journal of Water Resources Planning and Management*, 149(1):05022016, 2023. doi:10.1061/JWRMD5.WRENG-5589.
- [6] Arthur Drenoyanis, Raad Raad, I. Wady, and C. Krogh. Implementation of an IoT based radar sensor network for wastewater management. *Sensors*, 19(2):254, 2019. [\[Direct PDF\]](#) · doi:10.3390/s19020254.
- [7] Irja Helm, Gerli Karina, Lauri Jalukse, Todd Pagano, and Ivo Leito. Comparative validation of amperometric and optical analyzers of dissolved oxygen: a case study. *Environmental Monitoring and Assessment*, 190:313, 2018. [\[Direct PDF\]](#) · doi:10.1007/s10661-018-6692-5.
- [8] Mathieu Lepot, Andres Torres, Thomas Hofer, Nicolas Caradot, G"unter Gruber, Jean-Baptiste Aubin, and Jean-Luc Bertrand-Krajewski. Calibration of UV/Vis spectrophotometers: A review and comparison of different methods to estimate TSS and total and dissolved COD concentrations in sewers, WWTPs and rivers. *Water Research*, 101:519–534, 2016. [\[Direct PDF\]](#) · doi:10.1016/j.watres.2016.05.070.
- [9] Tianling Li, Melissa Winnel, Hao Lin, Jared Panther, Chang Liu, Roger O'Halloran, Kewen Wang, Taicheng An, Po Keung Wong, Shanqing Zhang, and Huijun Zhao. A reliable sewage quality abnormal event monitoring system. *Water Research*, 121:248–257, 2017. [\[Direct PDF\]](#) · doi:10.1016/j.watres.2017.05.040.
- [10] Gustavo A. Lopez-Ramirez and Alejandro Aragon-Zavala. Wireless sensor networks for water quality monitoring: A comprehensive review. *IEEE Access*, 11:95120–95142, 2023. [\[Direct PDF\]](#) · doi:10.1109/ACCESS.2023.3308905.
- [11] Saber Moradi, Tarek Zayed, and Farzad Golkhoo. Review on computer aided sewer pipeline defect detection and condition assessment. *Infrastructures*, 4(1):10, 2019. [\[Direct PDF\]](#) · doi:10.3390/infrastructures4010010.
- [12] Mario Pacheco Fern'andez, Danijel Despot, and Matthias Barjenbruch. Comparison of H₂S gas sensors: A sensor management procedure for sewer monitoring. *Sustainability*, 13(19):10779, 2021. [\[Direct PDF\]](#) · doi:10.3390/su131910779.
- [13] Sudhir Kumar Pandey, Ki-Hyun Kim, and Kea-Tiong Tang. A review of sensor-based methods for monitoring hydrogen sulfide. *TrAC Trends in Analytical Chemistry*, 32:87–99, 2012. [\[Direct PDF\]](#) · doi:10.1016/j.trac.2011.08.008.
- [14] Janna Prisutova, Anton Krynkina, Simon Tait, and Kirill Horoshenkov. Use of fibre-optic sensors for pipe condition and hydraulics measurements: A review. *CivilEng*, 3(1):85–113, 2022. [\[Direct PDF\]](#) · doi:10.3390/civileng3010006.
- [15] Chan H. See, Kirill V. Horoshenkov, M. T. Bin Ali, and Simon J. Tait. An acoustic sensor for combined sewer overflow screen condition monitoring in a drainage infrastructure. *Sensors*, 21(2):404, 2021. [\[Direct PDF\]](#) · doi:10.3390/s21020404.
- [16] B. Shi, S. Catsamas, B. Deletic, M. Wang, P. M. Bach, A. Lintern, A. Deletic, and D. T. McCarthy. Illicit discharge detection in stormwater drains using an Arduino-based low-cost sensor network. *Water Science and Technology*, 85(5):1372–1383, 2022. [\[Direct PDF\]](#) · doi:10.2166/wst.2022.034.
- [17] Andrzej Sikorski, Franklin Solano Donado, and Stanislaw Kozdrowski. Cost-efficient coverage of wastewater networks by IoT monitoring devices. *Sensors*, 22(18):6854, 2022. [\[Direct PDF\]](#) · doi:10.3390/s22186854.
- [18] Jayesh M. Sonawane, Chigozie I. Ezugwu, and Prakash C. Ghosh. Microbial fuel cell-based biological oxygen demand sensors for monitoring wastewater: State-of-the-art and practical applications. *ACS Sensors*, 5(8):2297–2316, 2020. [\[Direct PDF\]](#) · doi:10.1021/acssensors.0c01299.
- [19] Ee X. Tan, Anthony Anastasi, and Sudesh Chandra. Electrochemical detection of nitrate, nitrite and ammonium for on-site water quality monitoring. *Current Opinion in Electrochemistry*, 32:100926, 2022. [\[Direct PDF\]](#) · doi:10.1016/j.coelec.2021.100926.

- [20] U.S. Environmental Protection Agency. Combined sewer overflows: Guidance for monitoring and modeling. Technical Report EPA 832-B-99-002, Office of Wastewater Management, U.S. Environmental Protection Agency, Washington, DC, 1999. [\[Direct PDF\]](#).
- [21] Yelbek Utepov, Alexandr Neftissov, Timoth Mkilima, Assel Mukhamejanova, Shyngys Zharassov, Alizhan Kazkeyev, and Andrii Biloshchytskyi. Prototyping an integrated IoT-based real-time sewer monitoring system using low-power sensors. *Eastern-European Journal of Enterprise Technologies*, 3(5/123):6–23, 2023. [\[Direct PDF\]](#) · [doi:10.15587/1729-4061.2023.283393](https://doi.org/10.15587/1729-4061.2023.283393).
- [22] Yang Yu, A. Safari, X. Niu, Bruce Drinkwater, and Kirill V. Horoshenkov. Acoustic and ultrasonic techniques for defect detection and condition monitoring in water and sewerage pipes: A review. *Applied Acoustics*, 183:108282, 2021. [\[Direct PDF\]](#) · [doi:10.1016/j.apacoust.2021.108282](https://doi.org/10.1016/j.apacoust.2021.108282).